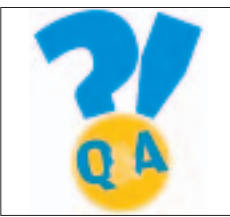




100 Bazaar



114 Agent 001

117 Your Questions,  
Our Answers

# Digital Tools

Technology For Personal And SoHo Productivity

## Main Story

We've all heard the hype created by Solid State Drives (SSDs). Touted as the magnetic storage or hard disk drive (HDDs) killers, SSDs represent something exciting; for it has been awhile since any breakthrough progress has been made in the storage domain. While processors have gone multi-core and graphics cards have grown ten times the number of shader units; hard drives have been making slow progress indeed. The obvious benefit of a mature technology is a proven track record and therefore reliability, which is more important for storage than speed. Hard drives have also gotten faster as manufacturers have pulled all the tricks they know by increasing areal density of the platters on an HDD, to increasing the buffer to even increasing the drives rpm. However, any mechanical device has limits to its performance increments and it seems hard drives have hit a solid wall; and any increments now, whether in reliability or performance are miniscule.

An SSD changes the rules of the game; storage is now not on magnetically oriented platters but on memory chips; either flash chips or RAM chips; the latter being much costlier in terms of price per unit of data stored (megabyte, gigabyte etc). Therefore an SSD is little more than a few memory chips on a PCB and a regular serial ATA interface. The most obvious benefit the lack of mechanical parts, which means less heating, increased performance (no mechanical delays), lower noise levels, and of course, less fragility. Also, since there is no read / write head, the operation is completed with minimal latency – unlike an HDD.

Therefore factors like file fragmentation do not matter. Most of the SSDs available today are flash-based. DRAM-based SSDs are very fast; and are bottlenecked by the SATA 2.0 bus' theoretical transfer limit of 300 MBps. They are also insanely expensive; which is why they are so rare.

**So, SSD or HDD?  
The two storage technologies one an old timer and survival expert; the other the cocky newcomer slug it out**

## War at the store

# SSD vs HDD



HD Tach (Intel SSD) – Note the very high read speeds and mediocre write speed

There are two main types of flash-based SSDs, those based on MLC (Multi-level cell) memory and those based on SLC (single-level cell) memory. Think of a cell as a single memory element; akin to a cell in our body. The former technology, MLC, has the benefit of being cheaper per gigabyte of storage. This is because each cell can store multiple bits of data; thus increasing storage density. This also means that errors are likely to occur, due to software complexities of such storage. SLC has the benefit of being much more reliable than MLC, since each cell only stores a single bit of data.

Flash drives have a couple of even more serious minuses. One; their write speeds are usually much slower than reads. Secondly, all flash memory has limited write (erase) cycles. For SLC, flash-memory cells can handle up to 1,00,000 writes; while for MLC this figure is much lower, in the region of 5,000 writes. Since some files like log files, file allocation tables and page files get written to very often (much more than flashes tolerance level) many consider flash to be unsuitable as yet for replacement of hard drives.

For our test, we used Intel's own 80 GB SSD drive for this comparison. The choice of hard drive was unanimous; Western Digital's new Velociraptor was our choice. This hard drive features a 2.5-inch form factor in a 3.5-inch chassis that helps reduce heating and a spindle rotational speed of 10,000 rpm. It supports SATA 2.0 and has a 16 MB buffer. This drive also offers greater storage than earlier Raptor hard drives at 300 GB.

HD Tach shows an interesting graph. While the read performance on the SSD drive is unbeatable and very close to the 300 MB per second theoretical limit that the SATA 2.0

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